



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO S.R.S FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

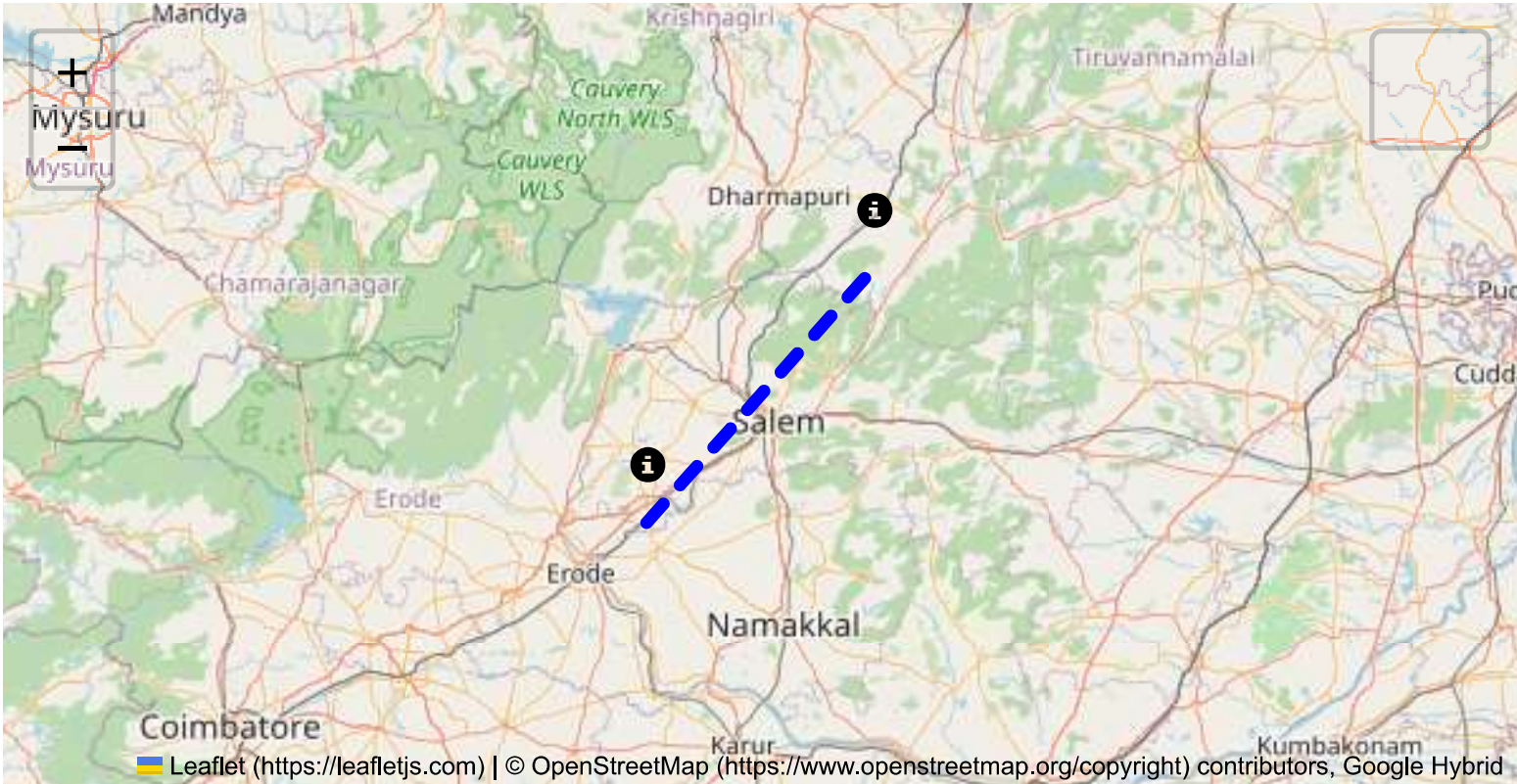
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

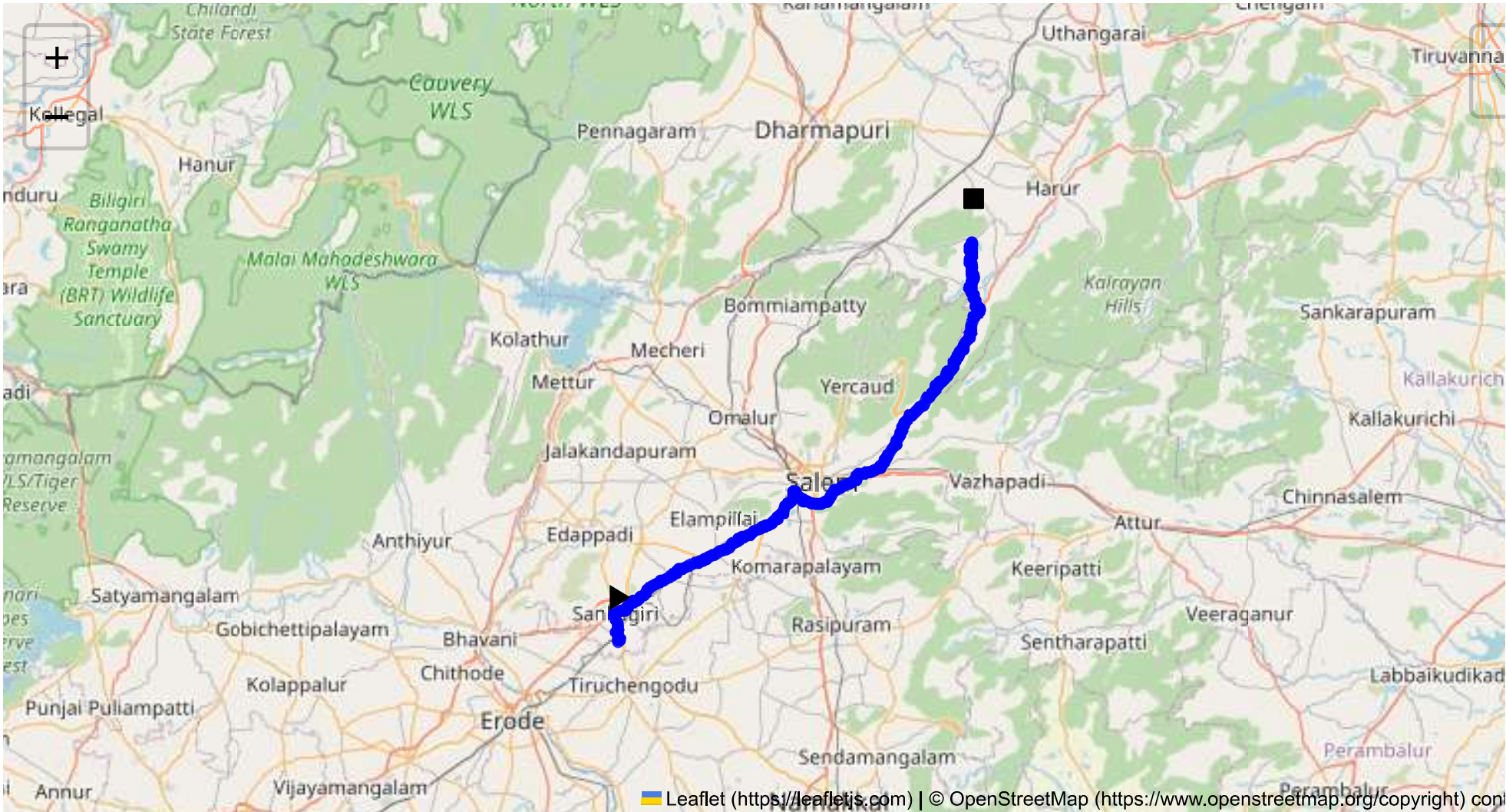
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 95.45 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.1 hours
Start: (11.4381, 77.8734)
End: (11.9806795, 78.3653729)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu to X9H8+V2C, Menasi Road, Pudinaattam, Tamil Nadu spans approximately 95.45 kilometers. This journey typically takes about 2.47 hours for heavy vehicles, including those carrying hazardous materials. The route primarily follows major roadways, which should be noted for their varying road conditions and traffic patterns.

Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with hot summers, a monsoon season, and mild winters. The monsoon, occurring between June and December, may cause heavy rains leading to poor road conditions and reduced visibility. Drivers should be especially cautious of flash flooding during this period as it could significantly hinder travel and safety.

Traffic Patterns and Congestion

While specific local traffic analysis is essential, typical congestion times in Tamil Nadu occur during morning (7:30 AM to 9:00 AM) and evening (5:00 PM to 7:00 PM) rush hours. Congestion-prone areas on this route may include major intersections and entry points to urban areas. It's advisable to plan travel outside these hours if possible to avoid delays.

Road Quality and Infrastructure

Road quality can vary significantly along the route. Some segments may suffer from potholes, narrow lanes, and a lack of shoulder space, particularly in rural stretches. Urban stretches typically offer better infrastructure but may have traffic management systems in place that could cause delays. Continuous monitoring of road maintenance updates can help anticipate potential roadblocks or detours.

Alternative Routes for Emergencies

In case of emergencies or blockages, smaller regional roads can serve as alternative paths. Routes parallel to the main itinerary should be mapped ahead of the journey. However, these routes may be less maintained and could require careful navigation, particularly for heavy vehicles.

Local Regulations on Hazardous Material Transport

Transporting hazardous materials is subject to strict regulations, including permitted travel times, specific routes, and safety protocols. Compliance with local and national regulatory bodies, such as the Petroleum and Explosives Safety Organisation (PESO), is mandatory. Regularly check for updates to ensure adherence to these regulations.

Historical Incidents

While specific historical incident data may not be readily available, typical issues include accidents caused by vehicle collisions and overturned trucks. Keeping abreast of local news and safety bulletins can provide insights into areas with frequent heavy-vehicle incidents.

Environmental Considerations and Sensitive Areas

The route crosses ecologically sensitive areas, particularly where agricultural activities dominate. Avoiding unnecessary stoppages and reducing speed can help mitigate environmental impact. Be aware of wildlife crossings and local flora and fauna to avoid any disruption.

Communication Coverage

Mobile and internet coverage can be spotty in rural areas along the route. While urban segments generally have strong signal coverage, it's essential to be prepared for dead zones by having alternative communication means like satellite phones or radio systems if necessary.

Estimated Emergency Response Times

Emergency response times can vary; urban areas might see responders arrive within 15-30 minutes, whereas rural areas could experience delays of up to an hour or more depending on accessibility and local infrastructure.

Overall Summary of Risk Assessment

The route has a moderate risk level for heavy vehicles transporting hazardous materials. Key risks include weather-related issues during the monsoon season, traffic congestion at peak times, and varying road conditions. Preparation, such as monitoring weather forecasts, planning around known congestion periods, adhering to regulations, and ensuring robust communication tools are in place, will help mitigate many potential hazards. Maintaining a vigilant approach will enhance safety and ensure compliance with local laws and environmental considerations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45000, 77.87213	30 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.53512, 77.95717	30 KM/Hr
13	Turn	Medium	11.53606, 77.95869	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	Medium	11.64140, 78.11801	30 KM/Hr
15	Turn	Medium	11.64179, 78.11822	30 KM/Hr
16	Turn	Medium	11.64218, 78.11881	30 KM/Hr
17	Turn	High	11.64192, 78.12133	15 KM/Hr
18	Turn	High	11.64116, 78.12168	15 KM/Hr
19	Turn	Medium	11.63934, 78.12082	30 KM/Hr
20	Turn	Medium	11.62873, 78.16541	30 KM/Hr
21	Turn	Medium	11.66372, 78.20649	30 KM/Hr
22	Turn	Medium	11.67130, 78.22990	30 KM/Hr
23	Blind Spot	Blind Spot	11.67111, 78.23134	10 KM/Hr
24	Turn	High	11.67203, 78.23152	15 KM/Hr
25	Turn	Medium	11.67445, 78.23920	30 KM/Hr
26	Turn	Medium	11.79051, 78.32169	30 KM/Hr
27	Turn	Medium	11.80448, 78.33329	30 KM/Hr
28	Turn	Medium	11.81243, 78.34087	30 KM/Hr
29	Turn	Medium	11.83483, 78.35295	30 KM/Hr
30	Turn	Medium	11.84746, 78.36069	30 KM/Hr
31	Turn	Medium	11.88930, 78.37653	30 KM/Hr
32	Blind Spot	Blind Spot	11.91963, 78.36214	10 KM/Hr
33	Turn	Medium	11.92065, 78.36272	30 KM/Hr
34	Turn	High	11.92149, 78.36217	15 KM/Hr
35	Turn	High	11.92719, 78.36526	15 KM/Hr
36	Turn	High	11.92717, 78.36596	15 KM/Hr
37	Turn	High	11.93567, 78.36849	15 KM/Hr
38	Turn	Medium	11.94314, 78.36668	30 KM/Hr
39	Turn	Medium	11.94348, 78.36643	30 KM/Hr
40	Turn	High	11.94952, 78.36650	15 KM/Hr
41	Turn	High	11.95040, 78.36594	15 KM/Hr
42	Turn	High	11.95100, 78.36621	15 KM/Hr
43	Turn	Medium	11.95307, 78.36410	30 KM/Hr
44	Turn	Medium	11.95399, 78.36235	30 KM/Hr
45	Turn	High	11.95446, 78.36218	15 KM/Hr
46	Turn	Medium	11.96807, 78.36511	30 KM/Hr
47	Turn	High	11.96948, 78.36348	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
48	Turn	Medium	11.97434, 78.36375	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
22	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
23	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
24	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
25	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
26	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
27	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
28	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
29	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
30	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
31	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
34	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
36	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
37	hospital	Pappireddipatti Government Hospital	11.9179195, 78.3646393	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium